

Group work set 2: Game Theory, Market Failure

Q1. Two firms, Arnold GmbH and Bobby GmbH, are both producing cars. They must each decide, simultaneously, between producing large cars and small cars. If both firms produce large cars, they each earn a profit of 5 Million EUR. If Arnold produces large cars and Bobby produces small cars, then Arnold earns 8 Million EUR, but Bobby earns only 2 Million EUR. If Bobby produces large cars and Arnold produces small cars, then Bobby earns 7 Million EUR and Arnold earns 3 Million EUR. If both produce small cars, they each earn 1 Million EUR.

- (a) Construct a payoff table.
- (b) For each of the cells in the table, state whether it is (1) Pareto-efficient and (2) Kaldor-Hicks efficient.
- (c) What is the Nash Equilibrium (or Nash Equilibria) of this game?
- (d) Suppose that Bobby moves first. Arnold observes Bobby's decision, and moves second. What is the Sub-game perfect Nash Equilibrium (or equilibria) of the game now?
- (e) Suppose Bobby and Arnold could negotiate at zero transaction cost. What would be the outcome of the negotiations according to the Coase Theorem?
- (f) Suppose that Arnold and Bobby meet a wizard. The wizard casts a spell on all consumers, causing them to see all cars as if they were the same size. That is, consumers can no longer differentiate between large cars and small cars. However, consumers still value large cars differently from small cars. Explain why this may lead to a new equilibrium, in which some types of cars (either large, or small, or both) are not sold.

Q2. Bill and George each own an ice cream stand. Each of them are trying to decide whether or not to offer Caramel-flavored ice cream. Suppose that their current revenues are 100\$ each, and that if both of them decide offer this flavor, then their revenues will not change. However, if one of them offers the new flavor and the other one does not, then the one who offers the flavor will increase his revenue to 160\$ and the other one will get a lower revenue of only 60\$. Assume that the cost of offering the Caramel flavor is 10\$.

- (a) What is the Nash Equilibrium of this game if they both decide simultaneously?
- (b) Is the Nash Equilibrium Pareto efficient? Is it Kaldor-Hicks efficient?
- (c) Compare this game to a Prisoner's dilemma. What is different and what is similar?
- (d) Suppose that Bill would decide first, and only then George will decide (after observing what Bill decides). What would be the Nash Equilibrium then?
- (e) Suppose that Bill and George can negotiate at zero cost. What would be the outcome of the game then? Would there be any money transfers between the parties?

Q3. Mary produces communication devices of two kinds: "A" and "B". A is worth 100 to consumers, whereas B is only worth 50. However, consumers cannot differentiate between A and B, because they both look exactly the same. Mary is the only producer in the market, and she expects consumers to want 50 products for a price of 20\$, but decides to produce only 15 products

and sell them for 40\$. Furthermore, the devices make a lot of noise when they are produced. One day, Mary asks a celebrity to recommend her devices, and suddenly everyone buys her devices.

Explain how each of the market failures covered in the course may manifest in market in which Mary operates in.